



Connect the motors and power the circuit with 12V battery. Now the circuit is ready for use. For instance, when you touch the forward arrow on the touchscreen, both the motors rotate in forward (clock-wise) direction simultaneously. Fit the motors properly with the wheels and the touchscreen near the hand to control the movement of the wheelchair.

The source program is written in 'C' language and compiled using AVR Studio to generate the Intel hex code. The generated hex code is programmed



The device is shipped with CKSEL="0001" and SUT="10". The default clock source setting is therefore 1MHz internal RC oscillator with longest startup time. To activate the internal oscillator, program the fuse bytes of the microcontroller as follows:

Fuse high byte = 99

The author is a final-year B.Tech in electronics & communications engineering. His interests include embedded systems and automation

The source code of this article is available for free download on www.efymag.com website.

```
#include<avr/io.h> //header file
#include<util/delay.h> //header file
void Drive_Motor(unsigned char)
LEFT,unsigned char RIGHT)//For MOTOR
{
    if(RIGHT==0)//if right == 0 then
right motor will stop
    {
        PORTB<=_BV(1);
        PORTB<=_BV(0);
    }
    if(RIGHT==1)//if right == 1 then
right motor will go forward
    {
        PORTB<=_BV(1);
        PORTB<=_BV(0);
    }
    if(RIGHT==2)//if right == 2 then
right motor will go backward
    {
        PORTB<=_BV(1);
        PORTB<=_BV(0);
    }
    if(LEFT==0)//if left == 0 then
left motor will stop
    {
        PORTB<=_BV(2);
        PORTB<=_BV(3);
    }
    if(LEFT==1)//if left == 1 then
left motor will go forward
    {
        PORTB<=_BV(2);
        PORTB<=_BV(3);
    }
}
```

```

PORTB&= ~BV(2);
PORTB|= _BV(3);
}
if(LEFT==2){//if left == 2 then
right motor will go backward
{
PORTB|= ~BV(2);
PORTB&= ~_BV(3);
}
}
void adc_init(void) //for adc initialization
{
ADCSRA=(1<<ADEN) | (1<<ADSC) | (1<<ADAFE) | (1<<
ADIFSC);
SFIOR=0x00;
unsigned char read_adc_channel(unsigned
char channel)//for channel selection
{
ADMUX=(1<<REFPS0) | (1<<ADLAR) | channel;
delay_ms(10);
return(ADCR);
}
int main(){// Main function
DDRB=0x0f;//for motor PB -{0,1,2,3}
connected to motor driver
adc_init();//adc initilized
unsigned char x,y;//for storing value of x
& y co-ordinate
while(1)

```

```
delay_ms(20);  
DRA=0x05; //00000101  
PORTA=0x01; //00000001for x init  
y=read_adc_channel(3);  
_delay_ms(20);  
delay_ms(20);  
DRA=0x0a; //00001010  
PORTA=0x08; //00001000//for y init  
y=read_adc_channel(2);  
_delay_ms(20);  
if(x>57 && x<140 && y>150)//forward  
{  
Drive_Motor(1,1); //PORTB=0b00001001;  
  
else if(x>57 && x<140 && y<60)//backward  
{  
Drive_Motor(2,2); //PORTB=0b00000110;  
  
} else if(x<70 && y>80 && y<130)//Left  
{  
Drive_Motor(2,1); //PORTB=0b00000101;  
  
} else if(x>150 && y>80 && y<130)//right  
{  
Drive_Motor(1,2); //PORTB=0b00000101;  
  
} else//stop  
{  
Drive_Motor(0,0); //PORTB=0b00000000;  

```